Gunshot Wound Classification via Transfer Learning and Kolmogorov-Arnold Networks

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***Abstract*—Classifying the directionality of a wound (entrance versus exit) and inferring medico-legal shooting distance (MLSD) constitute intricate forensic challenges that hinge on the judgment of trained specialists, and their outcomes are strongly influenced by lighting at the crime scene and the conditions under which photographs are captured. Even though machine learning (ML) has driven substantial progress in computer vision (CV), its use for these forensic problems has received comparatively little attention. Contemporary classifiers based on deep learning (DL) can already deliver strong results straight from unprocessed gunshot wound photographs. Nevertheless, published gunshot wound classification (GSWC) methods continue to fall short of the benchmarks reached in other CV application areas, including diabetic retinopathy detection, medical image diagnosis, and biometric recognition. To close this gap, we propose TransKAN, a hybrid framework that unites transfer learning with Kolmogorov-Arnold networks (KANs). TransKAN derives high-level spatial representations through a pre-trained ResNet152 backbone, and substitutes the standard fully connected classification head with a KAN, allowing it to capture complex non-linear relationships with greater efficiency. In order to evaluate TransKAN, we introduce the Gunshot Wound Image Database (GuWID), a collection of authentic photographic evidence drawn from homicide and suicide cases assessed by forensic specialists in incidents handled by the Federal District Civil Police (PCDF). Gathered from January 2012 through March 2022, GuWID contains 2,551 images (1,883 depicting entry wounds and 668 depicting exit wounds), classified by MLSD into 1,609 distant-range, 232 close-range, and 42 contact-range shots. Empirical findings demonstrate that TransKAN attains state-of-the-art (SOTA) results. GuWID dataset and source code can be accessed at https://github.com/Silva015/tcc-2026/tree/main/notebooks and https://github.com/pedrogarciafreitas/GuWID-UnB.**

***Index Terms*—Forensic image analysis, gunshot wound classification, transfer learning, Kolmogorov-Arnold networks, deep learning, medico-legal shooting distance**

I. Introduction

Handguns are the leading cause of homicides worldwide, causing three-quarters of killings in the Americas in 2021 [1]. Brazil records the most such deaths, surpassing 47,000 yearly [2]. Forensic professionals thus routinely examine gunshot wounds during crime scene work and autopsies. A key task involves distinguishing entry from exit wounds and classifying entry injuries by firing distance [3], [4].

Telling these wounds apart depends on assessing their shape, size, and edge features [5]. Yet these traits may be unclear, absent, or blended, making identification difficult. Bullet speed, angle of travel, ammunition kind, caliber, obstacles in the path, and the body area struck also change how a wound looks [6], [7]. These same factors also affect the legal classification of firing range, which uses visible tissue damage to sort wounds into contact, close-range, or distant groups [8]. Since this system relies solely on physical evidence on the body, it acts as a guide for reconstructing events [7]

To help courts remain impartial, forensic findings must be based on objective, scientifically proven standards [9]. ML and artificial intelligence (AI) have lately grown within forensic medicine to improve and speed up criminal probes [10], [11]. Computer vision has been applied to study ballistic injuries, but early studies were limited by small sample sizes, dependence on animal models [12], or failure to assess the MLSD when working with autopsy data [13].

Meanwhile, transformer-based classifiers have already reached competitive accuracy in other biometric domains, such as photoplethysmography-based health monitoring [14]–[17], underscoring how much room remains for similar gains in gunshot wound classification. Data-driven quality-descriptor advances have likewise extended to other imaging modalities, including geometry- and texture-aware quality assessment for 3D point clouds [18]–[20], further illustrating how broadly such descriptors are replacing traditional reference-based metrics. Recent studies show forensic practitioners increasingly adopting AI across their work, from digital forensics to forensic pathology [21]. Many see AI as a way to boost efficiency, accuracy, and automation in tasks like evidence analysis and case triage. For example, applying DL to forensic wound analysis could enable automated, precise, and repeatable testing [22], [23]. These studies mostly depend on traditional convolutional neural networks (CNNs), which use standard multilayer perceptrons (MLPs) in their final classification stages. Though these architectures work well under general conditions, they struggle to capture the highly non-linear, complex feature boundaries found in noisy, imbalanced, and unstandardized forensic wound images. Thus, more advanced classifiers are needed, ones able to extract and separate non-linear patterns without requiring vastly larger datasets or huge parameter counts.

To meet this need for classifiers that capture non-linear patterns without massive datasets or parameter counts, we